

CE 562: Numerical Methods for Engineering Computation

Course Code	CE 562
Course Name	Numerical Methods for Engineering Computation
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	Undergraduate (Elective) / Postgraduate
Mutual Exclusion	ME-504
Approval	59th BoA (Apr 2025)

Course Contents

- **Introduction:** Differential equations in engineering applications. Analytical vs. Numerical solution of a mathematical model. Computer representation of numbers. Errors in numerical computation. Review of CPP programming concepts - Program structure, data types, arrays, structures, functions, file handling using simple problems: Second moment of area, analysis of beams, basic statistics etc. [3 Lectures]
- **Systems of linear algebraic equation:** Gauss elimination, Gauss Jordan, LU decomposition, and Gauss-Seidel methods. Thomas algorithm for tri-diagonal and Cholesky decomposition for symmetric matrices. Matrix inversion methods. CPP programs for the mentioned schemes. [6 Lectures]
- **Eigen problem:** Eigenvalues and Eigen vectors. Properties. Methods of estimation of Eigenvalues and Eigenvectors – Power Iteration Methods, Jacobi Iteration, QR algorithm; Application of these concepts towards Matrix Inversion and solution of linear simultaneous equations. [5 Lectures]
- **Curve fitting:** Linear regression, polynomial regression, nonlinear regression. CPP program for constitution of normal equations. [4 Lectures]
- **Finite differences and interpolation:** Taylor's series, Forward, backward and central differences, Difference tables, Finite difference operators, Newton's forward and backward interpolation formulae, Stirling's, Bessel's and Laplace-Everett's interpolation formulae. Lagrange's polynomial and Newton's divided difference formula. CPP programs for implementing interpolation schemes. [6 Lectures]
- **Numerical differentiation and integration:** Derivatives using forward, backward and central difference formulae. Newton-Cotes integration formulae – Trapezoidal rule, Simpson's 1/3 rule, Simpson's 3/8 rule. CPP programs for implementing numerical integration schemes. [6 Lectures]
- **ODEs:** Description of initial and boundary value problems. Taylor's series, Picard's, Euler's, Runge-Kutta and Milne's methods. Finite difference solution of boundary value problems. [6 Lectures]
- **PDEs:** Description of Elliptic, Parabolic and Hyperbolic PDEs. Solution of Laplace's and Poisson's equations. Solution of linear heat transfer equation - CPP programs for implementation of explicit and implicit schemes. Conditions for stability. (6 Lectures)

Text Books

1. Steven C. Chapra and Raymond P. Canale, **Numerical methods for engineers**, McGraw- Hill, 2015.
2. Nita H. Shah, **Numerical methods with C++ programming**, PHI, 2009.

References

1. Joe D. Hoffman, **Numerical methods for engineers and scientists**, Marcel Dekker, 2001.
2. T. Veerarajan and T. Ramachandran, **Numerical methods with programs in C**, Tata McGraw-Hill, 2006.